

1. Use Case Name: Update Inventory

Textual Description:

The Store Manager updates the current inventory levels (shelf and warehouse quantities) for a specific product in the system. After updating the inventory, the system automatically checks if the product has fallen below its minimum stock threshold and triggers appropriate alerts and supplier orders if necessary.

List of Actors:

- Primary: Store Manager
- Secondary: System (performs automated checks)

Pre-conditions:

- The product exists in the database
- Inventory levels are defined for the product
- New quantities are valid (non-negative integers)

Post-conditions:

- Inventory levels are updated in the database
- System automatically executes "Check Minimum Stock"
- If stock is below minimum: Alert is generated and Supplier Order may be created

Main Success Scenario:

1. Store Manager selects "Update Inventory" from menu
2. System displays product selection interface
3. Store Manager enters Product ID
4. Store Manager enters new shelf quantity
5. Store Manager enters new warehouse quantity
6. System validates input (quantities ≥ 0)
7. System updates inventory in database
8. System automatically checks minimum stock threshold
9. System displays confirmation message
10. Use case ends successfully

Alternatives/Extensions:

- 1a: Invalid Product ID
 - At step 3: If product doesn't exist
 - System displays error message
 - Return to step 2
- 1b: Invalid Quantity Input

- At step 6: If quantities are negative
 - System displays error message
 - Allow user to re enter values
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- 1c: Automatic Alert Generation
 - At step 8: If new quantity < minimum threshold
 - System generates "Low Stock Alert"
 - Continues to step 10
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- 1d: Automatic Supplier Order
 - At step 8: If new quantity < minimum threshold
 - System automatically creates supplier order
 - Continues to step 10

2. Use Case Name:

Generate Alert (Low Stock Alert)

Textual Description:

The system automatically generates and records a low stock alert when a product's current inventory falls below its minimum threshold. The alert contains comprehensive information about the stock shortage and is made available to the Store Manager for review and action.

List of Actors:

- Primary: System (Automatic)
- Secondary: Store Manager (views alert)

Pre-conditions:

- Product quantity < minimum threshold
- Product exists in database

Post-conditions:

- Alert is visible in "View Low Stock Alerts"
- Notification may be sent to Store Manager

Main Success Scenario:

1. System detects: current_quantity < minimum_threshold
2. System saves Product to database (Alerts table)
3. System updates alert dashboard
4. Alert becomes visible in "View Low Stock Alerts"
5. System show notification to Store Manager
6. Use case ends successfully

Alternatives/Extensions:

- 2a: Alert Already Exists
 - At step 2: If alert for this product already exists
 - Doesn't create duplicate